AI LAB - SEMESTER PROJECT

RACE: Large-scale ReAdable Comprehension Dataset From Exams

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Instructor: AI Lab Faculty | **Focus:** Scikit-Learn NLP Pipeline

1. PROJECT OBJECTIVES

The core mission was to build an automated educational assessment tool that mimics human-level reasoning using classical ML.

- **Verification:** Identify the correct answer from 4 options.
- **Generation:** Produce realistic distractors (wrong answers).
- **Support:** Extract contextual hints from passages.

2. DATASET: THE RACE BENCHMARK

High Complexity

Unlike SQuAD, RACE requires reasoning across multiple sentences to find the answer.

Dual Levels

RACE-M: Middle School (Easier vocabulary).
RACE-H: High School (Complex reasoning).

Scale

~28,000 articles and nearly 100,000 questions total.

3. DATASET STATISTICS

Attribute	Mean Value
Article Length	321.9 words
Question Length	10.1 words
Option Length	6.4 words
Vocabulary Size	~140,000 unique tokens

The uniform distribution of labels (A, B, C, D) ensures no positional bias in the options, forcing the model to rely on semantic features rather than frequency.

4. EDA: SEMANTIC OVERLAP

[Visual: Histogram showing overlap distribution]

Key Finding:

Correct options tend to have a **32% higher** Jaccard Similarity with the passage compared to distractors.

However, "trap" questions often have high overlap in wrong options to fool simple keyword-matching models.

5. STATISTICAL ANALYSIS: K-MEANS

To see if the data is inherently separable without labels, we performed K-Means clustering (K=2).

74.7%

Purity Score

This high purity score indicates that the "Correct" vs "Incorrect" feature space is naturally distinct, allowing linear classifiers to perform well.

6. GMM & LABEL PROPAGATION

GMM

Gaussian Mixture Models were used to find the "soft" probability of an option being a distractor.

Label Prop

Tested if 10% labeled data could train the rest. Resulted in 43% F1 score.

Density

Confirmed that "Wrong" options cluster more tightly than "Correct" ones.

7. PREPROCESSING WORKFLOW

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1. JSON Parsing (extracting article, question, options)
2. Tokenization & Lowercasing (Regex-based)
3. Stopword Removal (Customized for educational text)
4. TF-IDF Vectorization (Max Features: 5,000)
5. Scaling (StandardScaler for lexical features)
```

8. FEATURE ENGINEERING

We engineered 5 critical non-textual features:

- **Cosine Similarity:** Option vs. Article vector.
- **Length Ratio:** Option length relative to the average option length.
- **Overlap Count:** Number of shared non-stopword tokens.
- **Position Score:** Where the keywords appear in the article.

9. MODEL SELECTION: VERIFICATION

We evaluated 4 models for the Answer Verification task:

1. **Logistic Regression** (Baseline)
2. **SVM** (High-dimensional focus)
3. **Random Forest** (Feature importance)
4. **Ensemble** (Weighted Average)

10. TRAINING PROCESS

Using a **80-20 Train/Test split** on the RACE-M subset:

- Implemented **GridSearchCV** for Logistic Regression regularization (C parameters).
- Utilized **Pipeline** objects to prevent data leakage during scaling.
- Training time: ~4 minutes on CPU.

11. PERFORMANCE: VERIFICATION

Model	Precision	Recall	F1-Score
Logistic Regression	0.48	0.45	0.46
SVM (Linear)	0.47	0.44	0.45
Weighted Ensemble	**0.49**	**0.47**	**0.48**

Note: Random chance on this 4-way classification is 25%.

12. DISTRACTOR GENERATION IMPLEMENTATION

Instead of generative AI (GPT), we used an **Extractive Ranking Approach**:

- Retrieve candidate sentences from the article using BM25-like scoring.
- Filter candidates that are too similar to the correct answer.
- Rank using our trained Logistic Regression model to find "believable" wrong answers.

13. EVALUATION: DISTRACTORS

0.007

BLEU Score

Low BLEU scores are expected in extractive distractor generation as the "gold" distractors in RACE are human-written and often paraphrased, while our system extracts from the text.

14. HINT EXTRACTION METHODOLOGY

Our backend extracts sentences that bridge the gap between the question and the answer:

```
- Sentence Similarity Ranking
- Keyword Density Filtering
- Threshold-based Selection
```

15. PERFORMANCE: HINTING

ROUGE-L

0.137

METEOR

0.112

Inference

< 1.0s

16. INTERPRETATION & CHALLENGES

Successes: The system identifies correct answers significantly better than random guessing without using heavy deep learning.

Limitations: Classical ML struggles with "negative" reasoning (e.g., "Which of the following is NOT true?") because it relies on positive keyword overlap.

17. SYSTEM DEMO

Running the Streamlit Backend

Watch as we input a custom passage and the system generates a full quiz with distractors and hints in real-time.